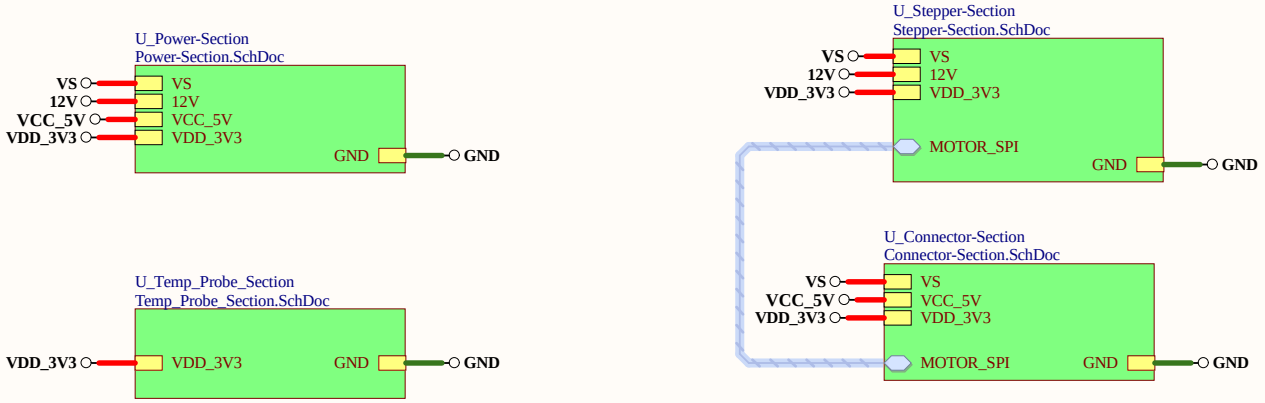
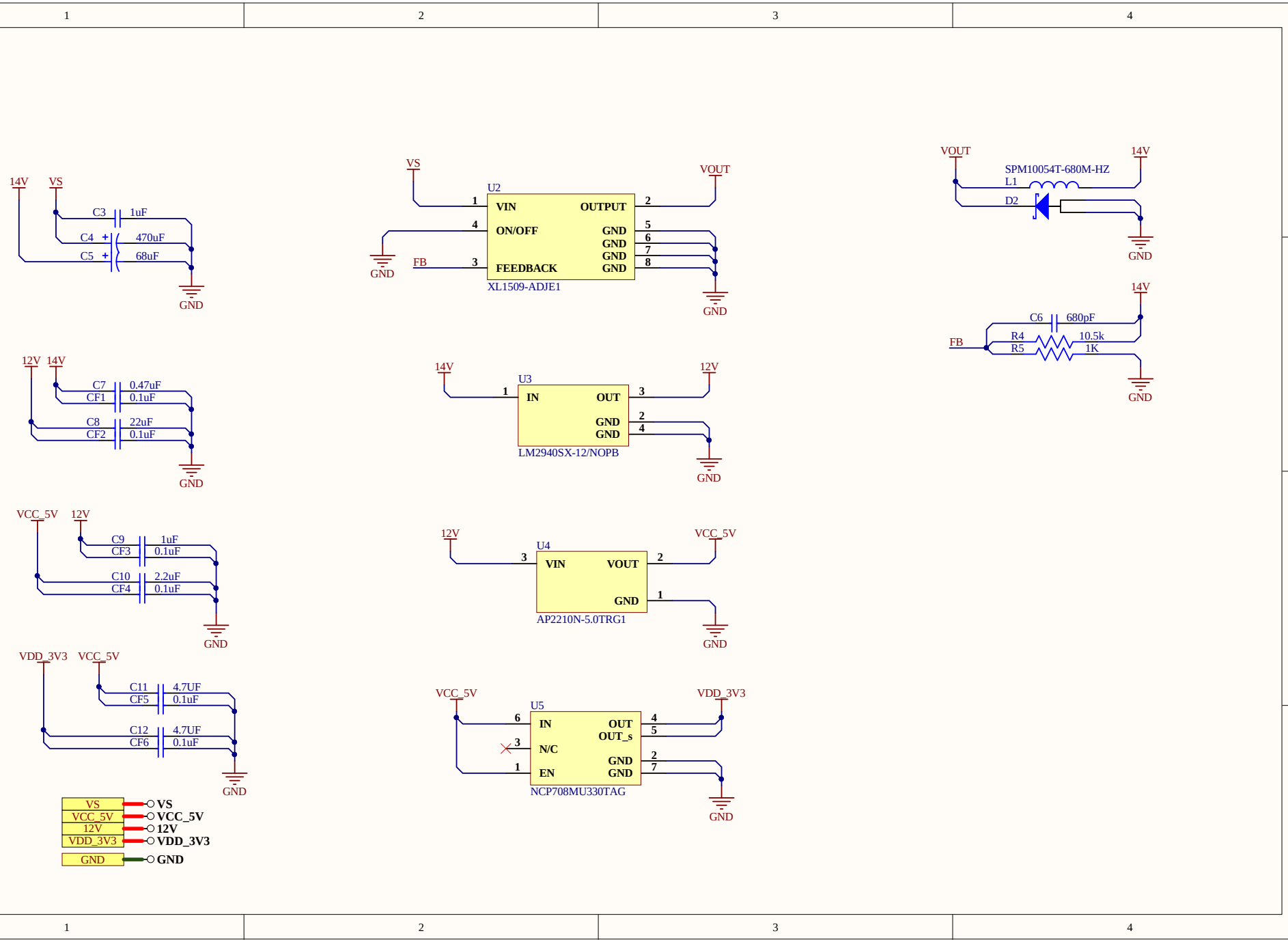
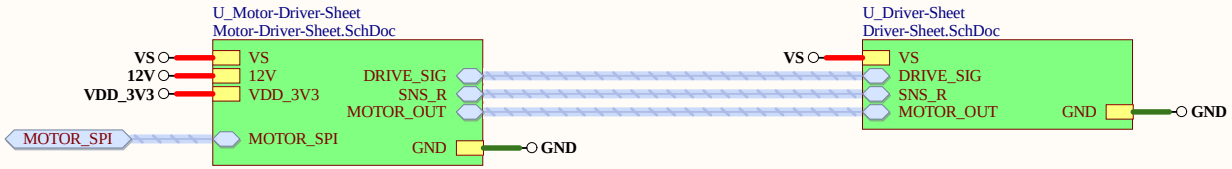
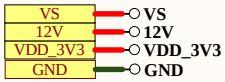
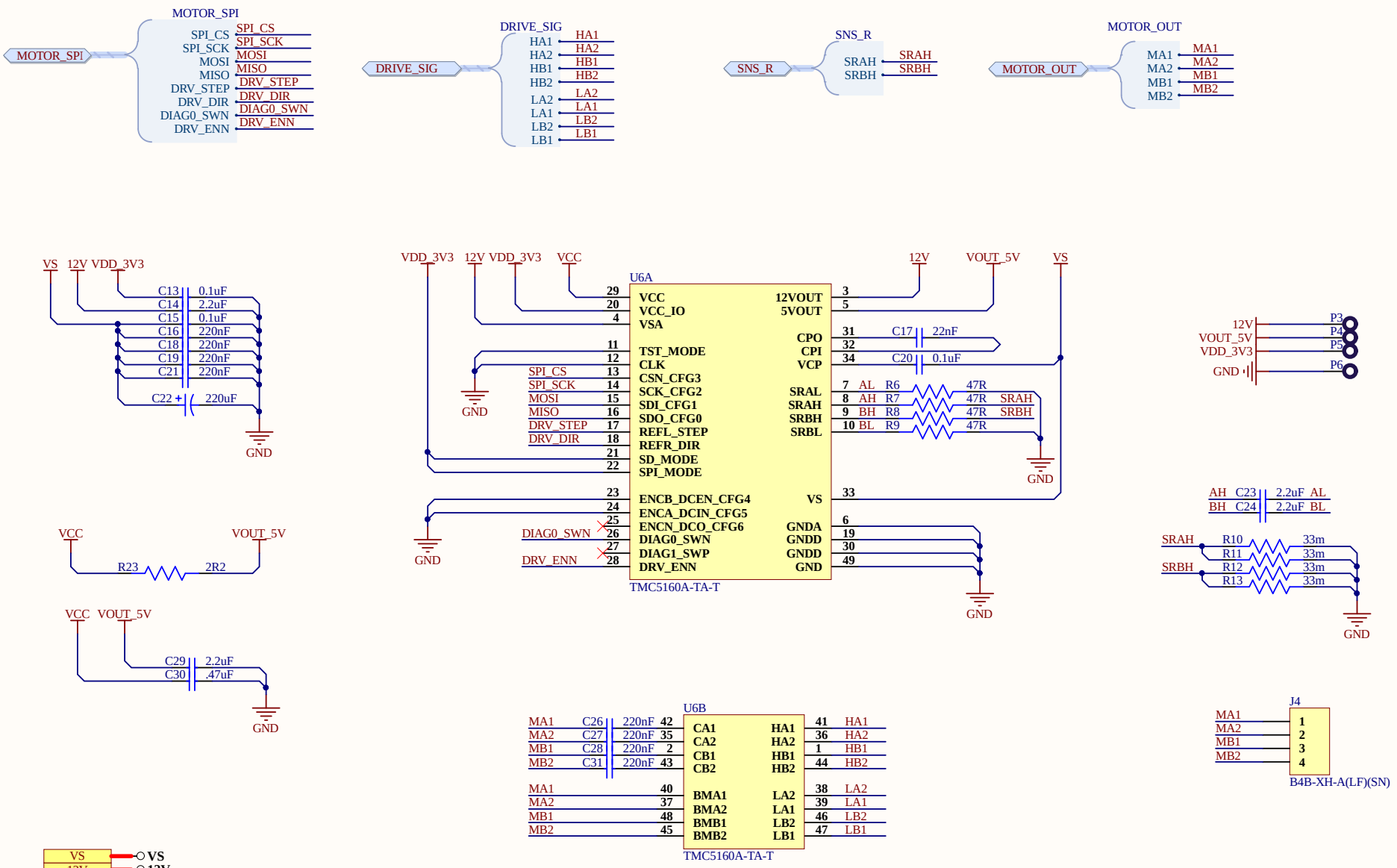
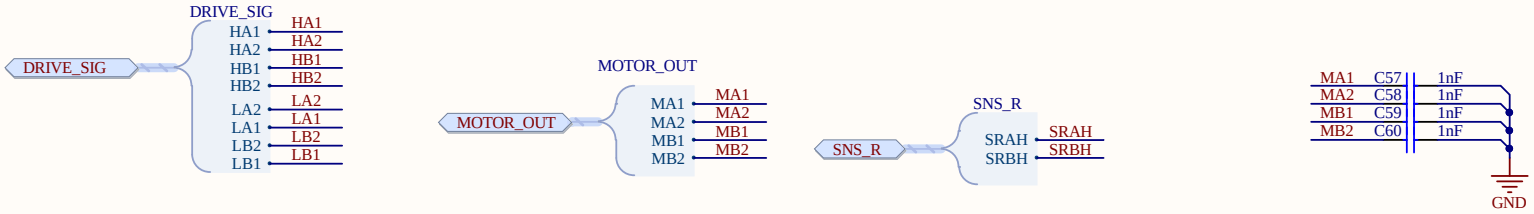

U_Power-Section
Power-Section.SchDoc
VS
12V
VCC_5V
VDD_3V3
VS
12V
VCC_5V
VDD_3V3
GND
GND
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
VDD_3V3
GND
GND
U_Stepper-Section
Stepper-Section.SchDoc
VS
12V
VDD_3V3
VS
12V
VDD_3V3
MOTOR_SPI
GND
GND
U_Connector-Section
Connector-Section.SchDoc
VS
VCC_5V
VDD_3V3
VS
VCC_5V
VDD_3V3
MOTOR_SPI
GND
GND

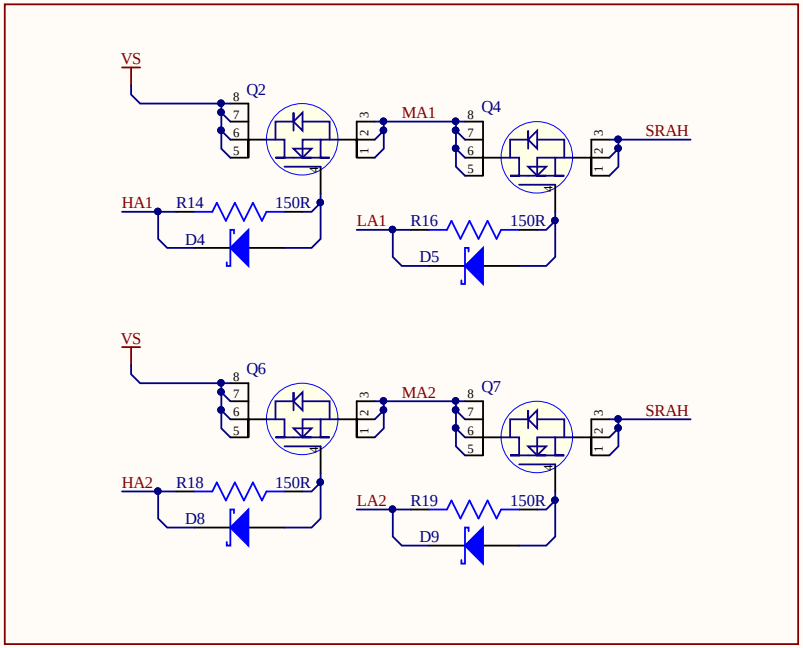

1
2
3
4
A
B
C
D
14V
VS
C3
1uF
C4
470uF
C5
68uF
GND
VS
U2
VIN
OUTPUT
ON/OFF
GND
FEEDBACK
XL1509-ADJE1
FB
VOUT
VOUT
14V
SPM10054T-680M-HZ
L1
D2
GND
14V
C6
680pF
R4
10.5k
R5
1K
FB
GND
12V
14V
C7
0.47uF
CF1
0.1uF
C8
22uF
CF2
0.1uF
GND
14V
U3
IN
OUT
GND
LM2940SX-12/NOPB
12V
GND
VCC_5V
12V
C9
1uF
CF3
0.1uF
C10
2.2uF
CF4
0.1uF
GND
12V
U4
VIN
VOUT
GND
AP2210N-5.0TRG1
VCC_5V
GND
VDD_3V3
VCC_5V
C11
4.7UF
CF5
0.1uF
C12
4.7UF
CF6
0.1uF
GND
VCC_5V
U5
IN
OUT
OUT_s
N/C
GND
EN
GND
NCP708MU330TAG
VDD_3V3
GND
VS
VCC_5V
12V
VDD_3V3
GND
VS
VCC_5V
12V
VDD_3V3
GND

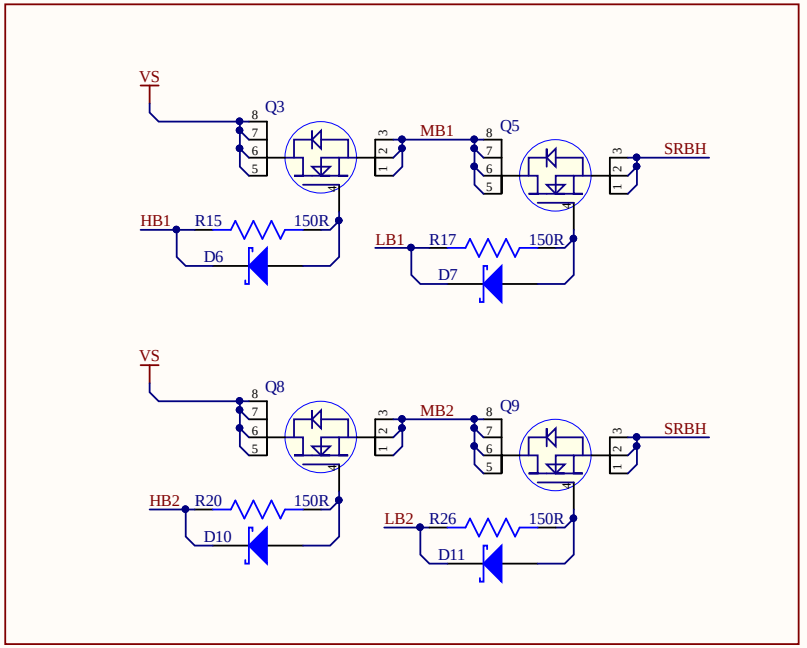

U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
12V
VDD_3V3
MOTOR_SPI
VS
12V
VDD_3V3
MOTOR_SPI
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND
U_Driver-Sheet
Driver-Sheet.SchDoc
VS
VS
DRIVE_SIG
SNS_R
MOTOR_OUT
GND
GND


VS
12V
VDD_3V3
GND
VS
12V
VDD_3V3
GND

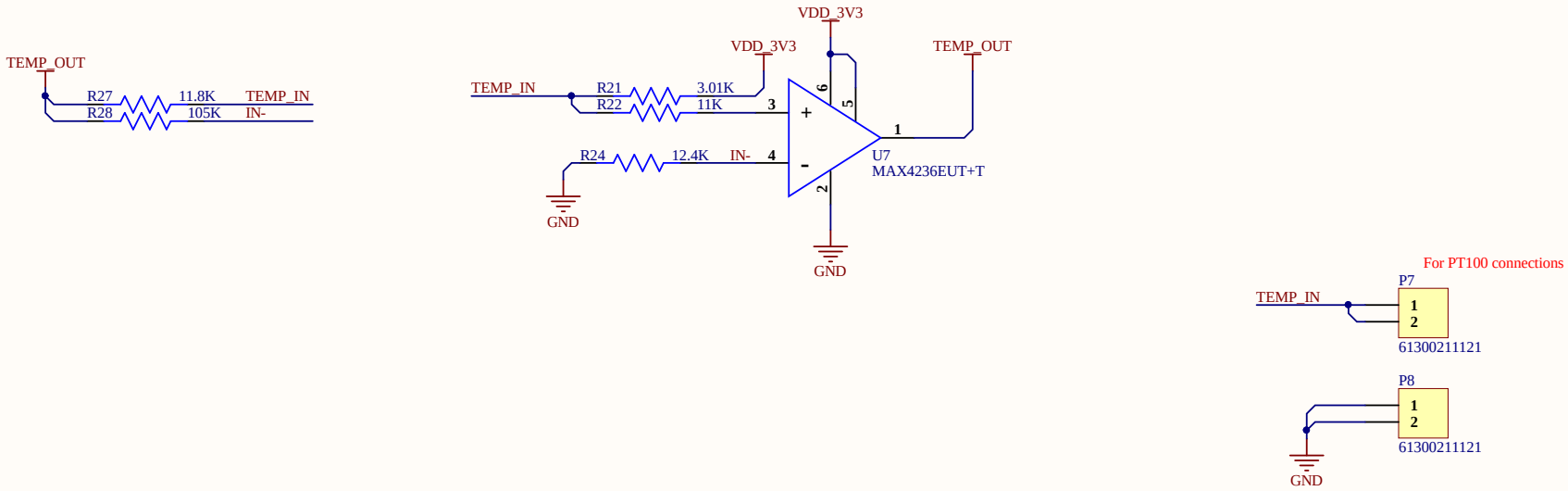

MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
SNS_R
SRAH
SRBH
MOTOR_OUT
MA1
MA2
MB1
MB2
VS
12V
VDD_3V3
C13 0.1uF
C14 2.2uF
C15 0.1uF
C16 220nF
C18 220nF
C19 220nF
C21 220nF
C22 220uF
GND
VCC
VOUT_5V
R23 2R2
C29 2.2uF
C30 .47uF
U6A
TMC5160A-TA-T
C17 22nF
C20 0.1uF
R6 R7 R8 R9 47R
C23 C24 2.2uF
R10 R11 R12 R13 33m
U6B
C26 C27 C28 C31 220nF
J4
B4B-XH-A(LF)(SN)
P3
P4
P5
P6

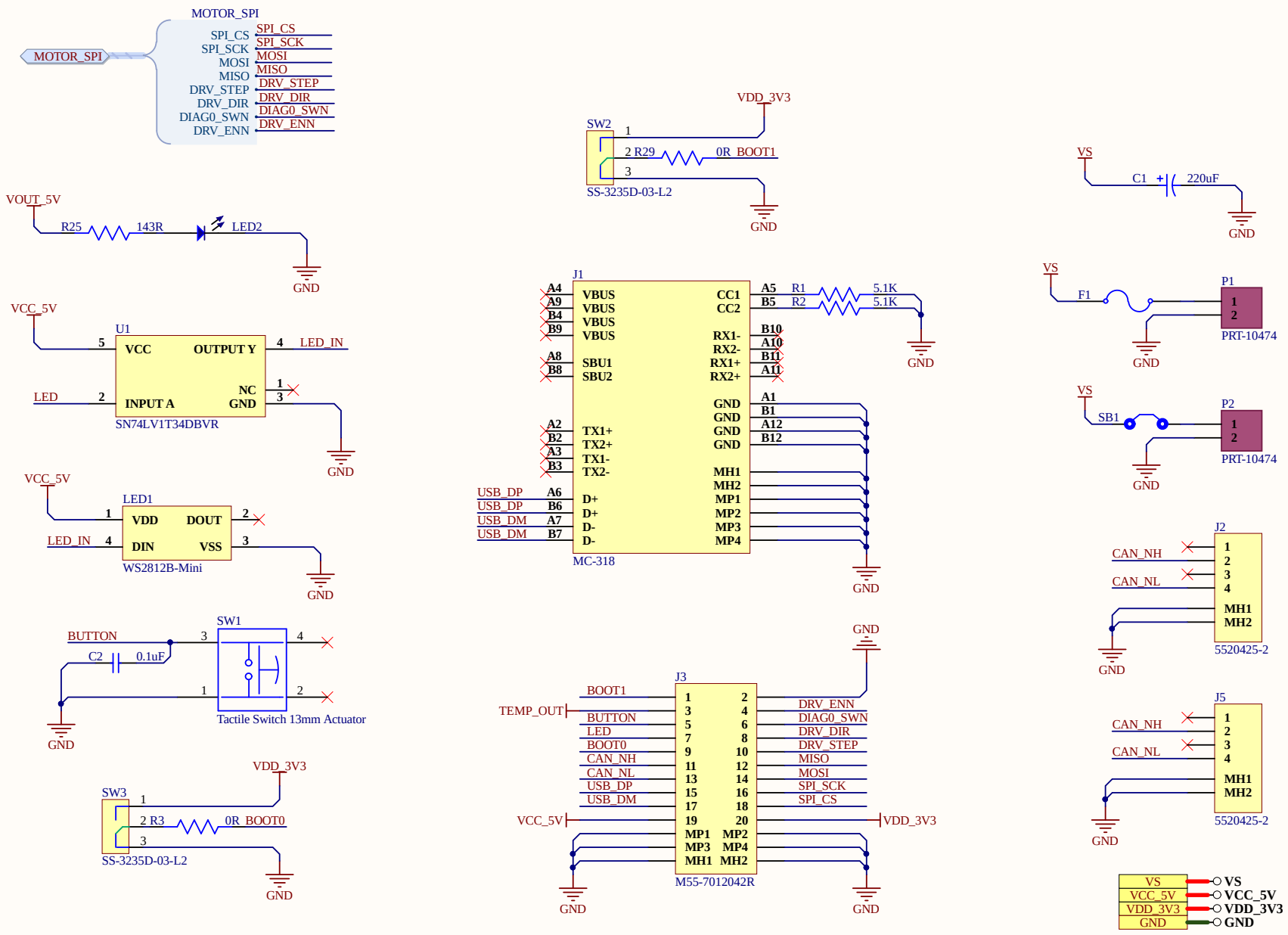

DRIVE_SIG
HA1
HA2
HB1
HB2
LA2
LA1
LB2
LB1
MOTOR_OUT
MA1
MA2
MB1
MB2
SNS_R
SRAH
SRBH
MA1 C57 1nF
MA2 C58 1nF
MB1 C59 1nF
MB2 C60 1nF
GND


VS
Q2
MA1
Q4
SRAH
HA1
R14
150R
D4
LA1
R16
150R
D5
VS
Q6
MA2
Q7
SRAH
HA2
R18
150R
D8
LA2
R19
150R
D9


VS
Q3
MB1
Q5
SRBH
HB1
R15
150R
D6
LB1
R17
150R
D7
VS
Q8
MB2
Q9
SRBH
HB2
R20
150R
D10
LB2
R26
150R
D11


VS
GND
VS
GND


TEMP_OUT
R27
11.8K
TEMP_IN
R28
105K
IN-
VDD_3V3
VDD_3V3
TEMP_OUT
TEMP_IN
R21
3.01K
R22
11K
3
6
5
+
1
R24
12.4K
IN-
4
-
U7
MAX4236EUT+T
2
GND
GND
For PT100 connections
TEMP_IN
P7
1
2
61300211121
P8
1
2
61300211121
GND
VDD_3V3
GND
VDD_3V3
GND


MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
VOUT_5V
R25
143R
LED2
GND
VCC_5V
U1
VCC
OUTPUT Y
LED_IN
LED
INPUT A
NC
GND
SN74LV1T34DBVR
LED1
VDD
DOUT
DIN
VSS
WS2812B-Mini
BUTTON
C2
0.1uF
SW1
Tactile Switch 13mm Actuator
SW3
2 R3
0R_BOOT0
SS-3235D-03-L2
VDD_3V3
SW2
2 R29
0R_BOOT1
J1
VBUS
SBU1
SBU2
CC1
CC2
RX1-
RX2-
RX1+
RX2+
TX1+
TX2+
TX1-
TX2-
D+
D-
MH1
MH2
MP1
MP2
MP3
MP4
R1
R2
5.1K
MC-318
J3
BOOT1
TEMP_OUT
BUTTON
LED
BOOT0
CAN_NH
CAN_NL
USB_DP
USB_DM
DRV_ENN
DIAG0_SWN
DRV_DIR
DRV_STEP
MISO
MOSI
SPI_SCK
SPI_CS
VCC_5V
VDD_3V3
M55-7012042R
VS
C1
220uF
F1
P1
P2
SB1
PRT-10474
J2
J5
5520425-2
VS
VCC_5V
VDD_3V3
GND